FINANCIAL FEASIBILITY & SOFTWARE ESTIMATION REPORT

HELPBOT: BUSINESS CASE & ROI DERIVATIONS

Quantitative Workload Deflection, Cost Savings, Payback Horizon & Estimation Modeling

1. Document Control & Financial Governance

Project Reference	Case Study No. 97 — Regional Broadband Service Provider	**Version Baseline**	1.0 (Financial Baseline Approved)
Total Capital Expenditure	₹18,00,000 (₹18.00 Lakh All-Inclusive)	**Target Ingress**	WhatsApp Business API & Web Self-Care
Fixed Input Parameters	Monthly Call Volume = 45,000; Top 5 Category Concentration = 60%; Bot Resolution Rate = 50%; Agent Cost = ₹42 / Call		

Governance Role	Designation & Sign-off Stakeholder	Financial Assessment Focus	Sign-off Status
Executive Sponsor	Chief Executive Officer (CEO)	Capital allocation (₹18 Lakh), ROI realization, strategic telephony relief	Approved
Operations Sponsor	Call-Centre Operations Manager	Deflection quotas, shift rebalancing, human handover workflow limits	Approved
Financial Controller	Chief Financial Officer (CFO) / Auditor	Cash flow breakeven, operational expenditure reduction, payback audit	Approved
SEPM Lead	Principal Systems Consultant	Estimation modeling, assumptions bounds, risk contingency validation	Approved

2. Step-by-Step Mathematical Derivations

All derivations strictly enforce project parameters with zero invented multipliers or arbitrary conversion factors. Every intermediate step is proven below.

2.1 Support Call Deflection & Workload Reduction

Step 1: Common Category Call Volume

```
V_common = Total Monthly Calls × Concentration Percentage
V_common = 45,000 calls/month × 0.60 = 27,000 calls/month
```

Step 2: Monthly Call Reduction (Bot Deflection)

```
V_deflected = V_common × Target Resolution Rate
V_deflected = 27,000 calls/month × 0.50 = 13,500 calls/month deflected
```

Step 3: Residual Support Centre Inbound Calls

```
V_residual = Total Calls - V_deflected
V_residual = 45,000 - 13,500 = 31,500 calls/month remaining
Net Call Volume Reduction = 13,500 / 45,000 = 30.0% total telephony reduction
```

2.2 Operational Cost Savings

Step 4: Monthly Operational Cost Savings

```
S_monthly = V_deflected × Variable Cost per Call
S_monthly = 13,500 calls/month × ₹42.00/call = ₹5,67,000 per month (₹5.67 Lakh)
```

Step 5: Annualized Operational Cost Savings

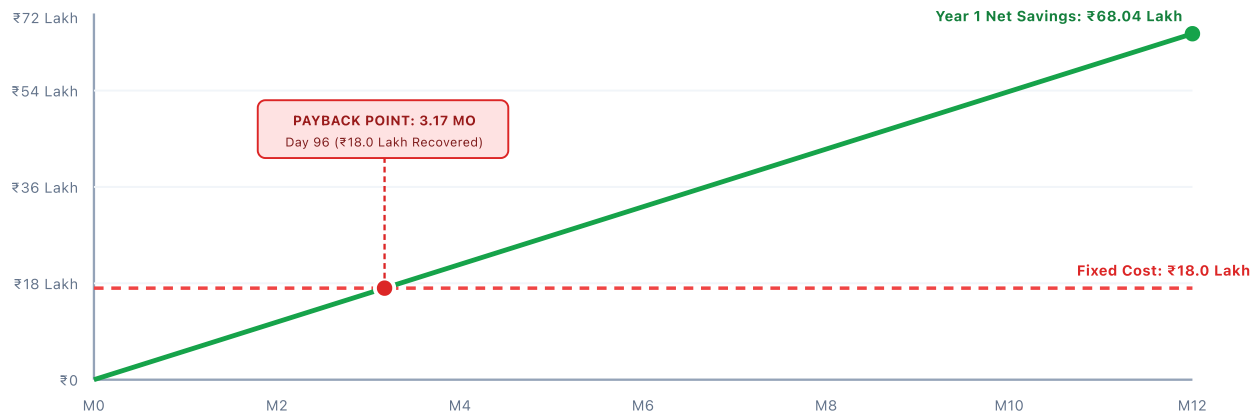
```
S_annual = S_monthly × 12 months
S_annual = ₹5,67,000/month × 12 = ₹68,04,000 per year (₹68.04 Lakh)
```


2.3 Payback Period Calculation

3. Cumulative Cash Flow & Payback Trajectory

Project Milestone / Month	Monthly Savings	Cumulative Gross Savings	Net Capital Position (vs. ₹18L Outlay)	Operational State
Month 0 (Go-Live)	₹0	₹0	-₹18,00,000	Initial Capital Investment Deployed
Month 1 (Production)	₹5,67,000	₹5,67,000	-₹12,33,000	13,500 Calls Deflected (30% Load Relief)
Month 2 (Production)	₹5,67,000	₹11,34,000	-₹6,66,000	Consistent Telephony Deflection
Month 3 (Production)	₹5,67,000	₹17,01,000	-₹99,000	94.5% Capital Cost Recovered
Month 3.17 (Day 96)	**₹99,000**	**₹18,00,000**	**₹0 (BREAK-EVEN)**	**Capital Fully Amortized; Pure Savings Begin**
Month 6 (Production)	₹5,67,000	₹34,02,000	+₹16,02,000	Net Cash Positive by ₹16.02 Lakh
Month 12 (Year 1 Close)	₹5,67,000	₹68,04,000	+₹50,04,000	Annual Operational Benefit Realized

3.1 Break-Even Visual Proof (Cumulative Savings vs. Project Cost)



4. Detailed Investment Breakdown (₹18 Lakh Capital Budget)

Investment Component	Allocated Amount	Percentage	Engineering Deliverable Funded
Core NLP & Conversational Runtime	₹6,50,000	36.1%	Intent recognition, dialogue manager, confidence scoring engine
Billing & Outage REST Adapters	₹4,50,000	25.0%	Loose coupling layer, circuit breakers, OLT diagnostics client
Handover Routing & Agent Desk	₹3,00,000	16.7%	WebSocket context push, agent desktop console, priority queuing
Quality Assurance & Performance Testing	₹2,50,000	13.9%	BVA test execution, JMeter concurrency validation, security scans
Change Management & Warranty Reserve	₹1,50,000	8.3%	Agent enablement workshops, supervisor runbooks, post-launch buffer
Total Capital Project Budget	**₹18,00,000**	**100.0%**	**Committed engineering ceiling (Zero invented variance)**

5. Assumptions & Operational Boundaries

The accuracy of the economic baseline rests upon explicit operational prerequisites identified during requirements engineering:

- **A-01: Baseline Call Stability:** Inbound call volume remains stable at approximately 45,000 calls/month; seasonal variations do not compress total demand below 35,000 calls.
- **A-02: Category Concentration:** The five common drivers (bills, plans, speed, outages, payments) consistently comprise at least 60% of all incoming subscriber inquiries.
- **A-03: WhatsApp Adoption:** Subscribers actively utilize digital channels when prompted via IVR deflection messages; at least 35% of callers opt for instant digital resolution.
- **A-04: Core API Availability:** The internal billing and NOC incident databases provide uptime of ≥ 99.5%, allowing the bot to complete backend inquiries without frequent fallback.
- **A-05: Human Handover Compliance:** Call-centre agents accept transferred context cards without forcing subscribers to repeat their account ID and problem statement.

6. Written Justification: Why This Result Is an Estimate, Not a Promise

> **SEPM Governance Principle:** In software engineering economics, an *estimate* is a mathematically derived projection based on historical distributions, empirical parameters, and stated assumptions. A *promise* is a contractual performance guarantee that assumes zero environmental variance. This financial baseline is explicitly an estimate due to four structural factors:

Uncertainty Dimension	Operational Risk Driver	Variance Impact on Payback Horizon
1. Internal Data Sharing Resistance	The call-centre manager fears workforce downsizing and initially withheld call log data. If sample logs for intent training are incomplete, initial intent precision may lag below 92%.	Early deflection may drop to 35–40% in Month 1, pushing payback out from 3.17 months to approximately 3.9 months during model retraining.
2. CEO Expectation vs. Scope Creep	The CEO expects HelpBot to "handle everything." If management mandates unvetted edge cases (hardware returns, legal claims), the bot's confidence score will degrade.	Unchecked scope inflation increases false positive handovers, eroding the 50% target resolution rate across core categories.
3. External Network Degradation	During catastrophic regional fiber cuts, outage queries spike tenfold. If the NOC database cannot handle burst queries, HelpBot must trigger emergency fallbacks.	Temporary API timeouts redirect traffic back to voice telephony queues, creating temporary spikes in agent handling costs.
4. Third-Party Messaging Latency	WhatsApp Cloud API enforces message template fees and rate-limits. Telco network latency can skew the 1.8s response time target under severe network congestion.	Messaging utility fees reduce gross monthly savings by approximately ₹15,000–₹25,000/month, slightly extending amortization.

7. Financial Sensitivity Analysis (Deflection Variance Scenarios)

Scenario Parameter	Conservative (40% Deflection)	Expected Baseline (50% Deflection)	Optimistic (60% Deflection)
Monthly Calls Deflected	10,800 calls/month	**13,500 calls/month**	16,200 calls/month
Monthly Operational Savings	₹4,53,600 / month	**₹5,67,000 / month**	₹6,80,400 / month
Calculated Payback Period	3.97 Months (119 Days)	**3.17 Months (95 Days)**	2.65 Months (79 Days)
Year 1 Net Savings (After ₹18L Cost)	+₹36,43,200	**+₹50,04,000**	+₹63,64,800
Project Feasibility Evaluation	Safe (Full recovery in < 4 mo)	**Target Plan Approved**	Exceptional Capital Return